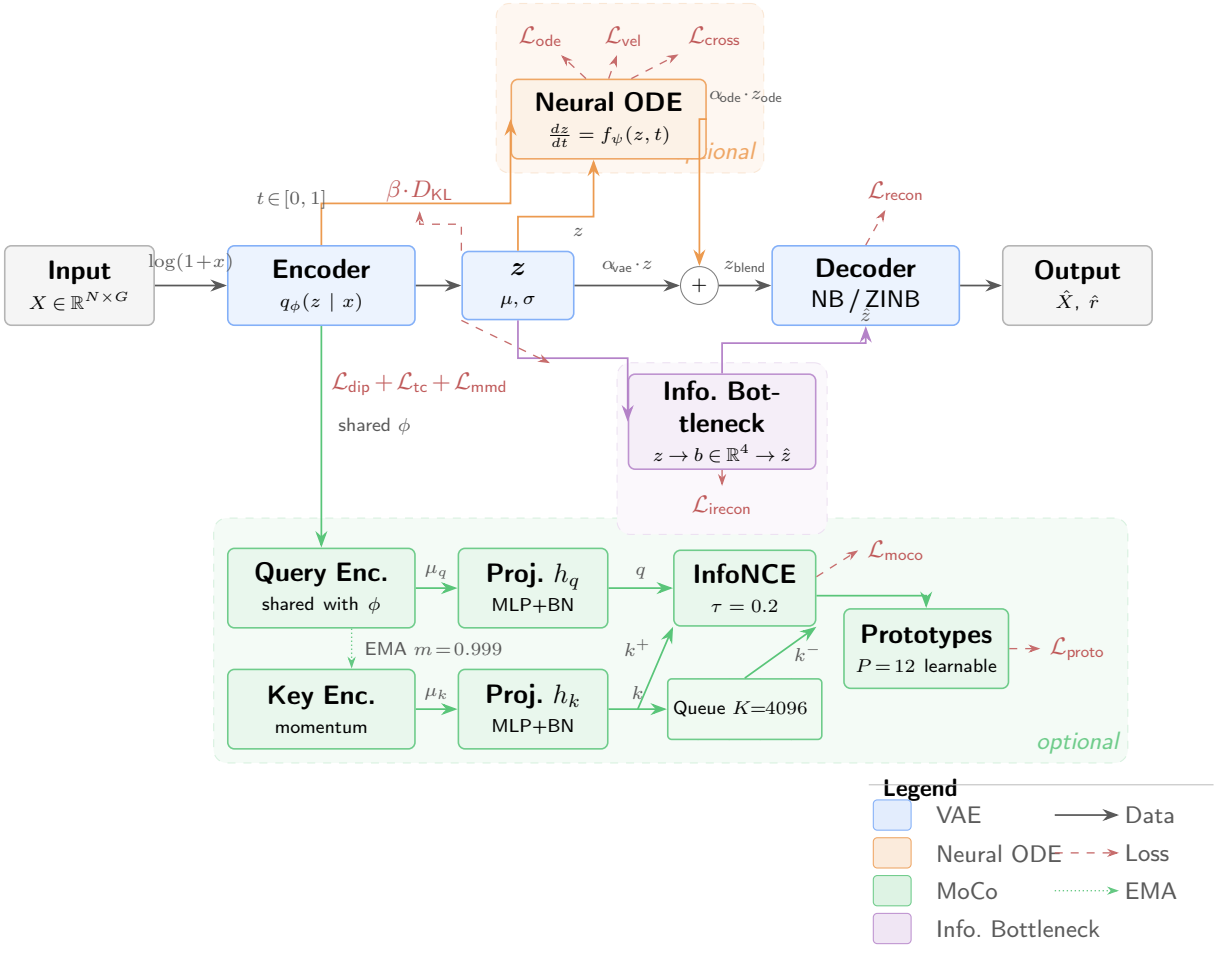

$\mathcal{L}_{\text{ode}}$
$\mathcal{L}_{\text{vel}}$
$\mathcal{L}_{\text{cross}}$
Neural ODE
$\frac{dz}{dt} = f_\psi(z, t)$
$\alpha_{\text{ode}} \cdot z_{\text{ode}}$
ional
$t \in [0, 1]$
$\beta \cdot D_{\text{KL}}$
z
$\mathcal{L}_{\text{recon}}$
Input
$X \in \mathbb{R}^{N \times G}$
$\log(1+x)$
Encoder
$q_\phi(z \mid x)$
z
μ, σ
$\alpha_{\text{vae}} \cdot z$
$+$
z_{blend}
Decoder
NB / ZINB
$\hat{z}$
Output
$\hat{X}$, $\hat{r}$
$\mathcal{L}_{\text{dip}} + \mathcal{L}_{\text{tc}} + \mathcal{L}_{\text{mmd}}$
shared ϕ
Info. Bottleneck
$z \to b \in \mathbb{R}^4 \to \hat{z}$
$\mathcal{L}_{\text{irecon}}$
Query Enc.
shared with ϕ
μ_q
Proj. h_q
MLP+BN
q
InfoNCE
$\tau = 0.2$
$\mathcal{L}_{\text{moco}}$
Prototypes
$P = 12$ learnable
$\mathcal{L}_{\text{proto}}$
EMA $m = 0.999$
k^+
k^-
Key Enc.
momentum
μ_k
Proj. h_k
MLP+BN
k
Queue K=4096
optional
Legend
VAE
Data
Neural ODE
Loss
MoCo
EMA
Info. Bottleneck